

Experimenting With Solar Power

Experience with me the fun of seeing a small solar panel run a cordless phone and the ampli-speaker box of a desktop PC

■ SOMNATH BERA

Friends call me impractical but I listen to my heart and that's it. So I purchased this 10W, 12V solar panel and on one Saturday afternoon me and my daughter erected it on the parapet wall of our house, 18 degrees inclined to north (as the latitude of Mumbai is 18 degrees North and 72 degrees East). We toiled till evening to get it properly fixed on the wall. The next day, i.e., Sunday, was the big day for us. My daughter and me got up and sat ready from 5 am in the morning.

The sun rose at 5:35 am. But the first glow of the bulb connected to the panel was seen at 6:55 am. More power started coming as the sun rose over the sky. At 9 am, the panel was delivering almost 9.5 watts of power at 12 volts. We rejoiced over cups of tea.

At 9:45 am, the output was 10.15 watts. My daughter perused me to replace the power supply of the ampli-speaker box of our desktop PC which runs on a step-down 12V, 500mA supply. It worked nicely. In the similar way, we replaced the 12V power supply adaptor of our cordless telephone with the solar power and it worked flawlessly. We tried to recharge our laptop too but it did not, as it requires 3A, 19 volts.

Next we tried to recharge our Nokia cellphone. Cellphones normally require 5V supply. The panel being 12V, would have blown the cellphone off. So we stepped it down to 5 volts with the help of a 3-pin regulator IC (IC 7805) very easily. But this is not



10W, 12V solar panel mounted on the parapet wall of my house. Typical data for Mumbai: latitude 18 degrees north, 72 degrees east; normal maximum daylight 4-5 hours/day

an efficient process and it's better to use smaller-capacity solar panels having 5V regulated output (viz, the output stabilised by a 5V zener diode, something in the form of shunt regulator). Remember a solar panel does not damage on its own if the output is short-circuited.

The 10W power was available till 3 pm. Then it went on reducing gradually till 6 pm when it became completely dead. In the whole day, we received little more than 75 watt-hours of power from the solar panel.

This left me wondering why cell-

phone manufacturers don't integrate a small solar panel on the back of the phone itself. For charging the phone, just put the phone out in the sun, thereby having more convenience and utility.

Next we coupled the solar panel with a 20Ah, 12V battery (generally used in motorcycles). The panel was then ready for work at night also.

Following devices can be safely run on this solar panel setup:

1. Cordless phone (runs 24 hours)
2. Small digital clock (LED type runs 24 hours)
3. Set-top box of the TV (runs 4-5 hours in a day)

It's really a fun seeing the panel work.

Considering the 20 per cent efficiency of a typical solar photovoltaic panel, if a small towel-size panel can produce so much power, just imagine the potential of solar power left unutilised.

The good news is that the cost of solar panels is reducing. At present, it is Rs 160 per watt here in Mumbai. ●

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